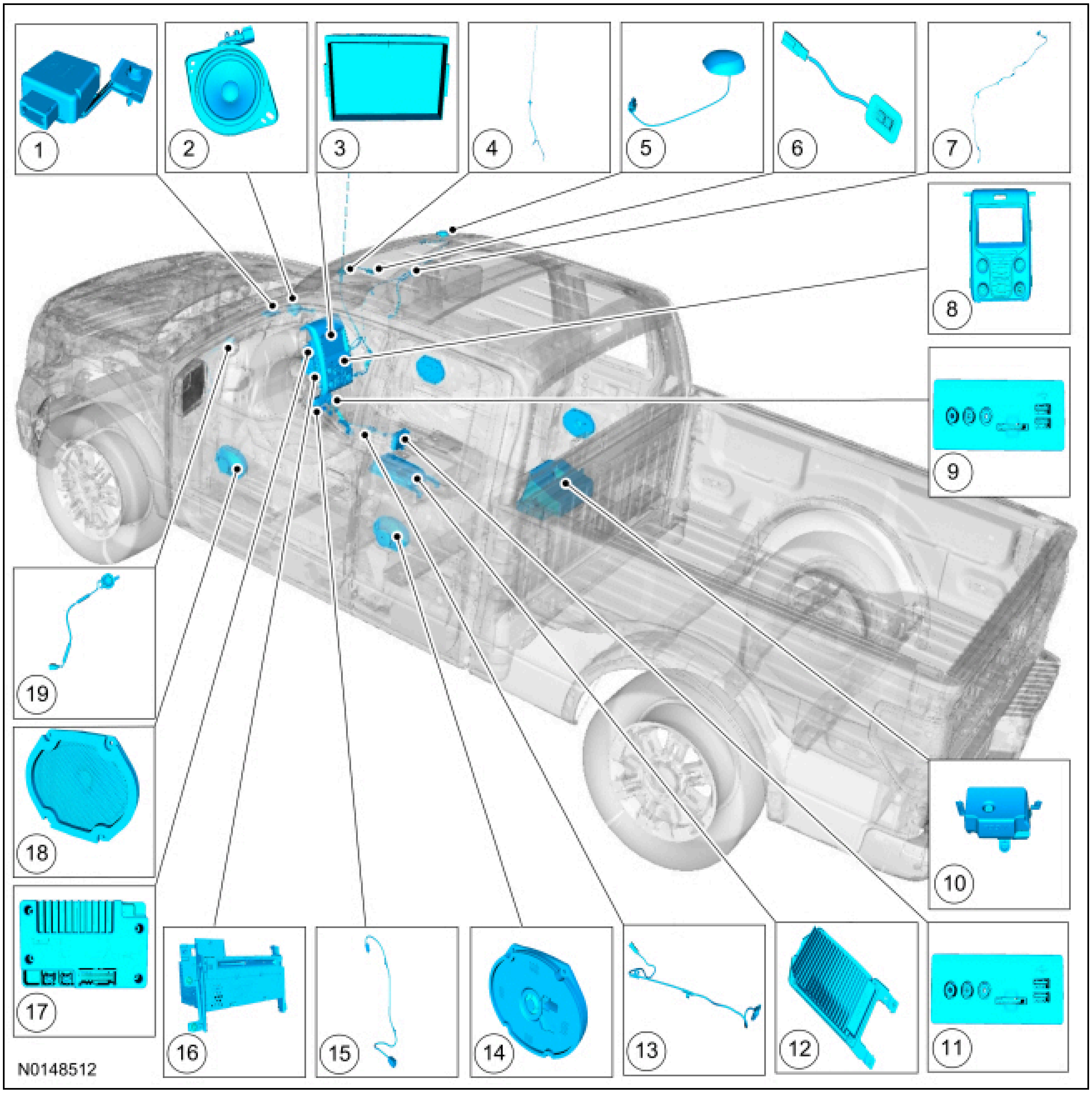
# Information and Entertainment System

Ford Ecat Electronic Parts Catalog

### Component Location




| Item | Description | Comments |
| --- | --- | --- |
| 1 | GPSM | Located in the top center area of the instrument panel |
| 2 | Instrument panel speaker | - |

| Item | Description | Comments |
|---|---|---|
| 3 | FDIM | Attached to the APIM |
| 4 | AM/FM/HD Radio® antenna and AM/FM/HD Radio® coaxial cable | HD Radio® (if equipped) |
| 5 | Satellite radio antenna | If equipped |
| 6 | SYNC® microphone | Located in the headliner |
| 7 | Satellite radio antenna coaxial cable | If equipped |
| 8 | FCIM | - |
| 9 | Media hub | Located in the instrument panel (with split bench) |
| 10 | Subwoofer speaker enclosure | Located under the LR seat |
| 11 | Media hub | Located in the console (with front captain chairs) |
| 12 | Audio DSP module | With split bench shown, with front captain chairs similar |
| 13 | USB cable | Located in the console (with front captain chairs) |
| 14 | Rear door speaker | - |
| 15 | USB cable | Located in the instrument panel (with split bench) |
| 16 | ACM | - |
| 17 | APIM | Attached to the FDIM |
| 18 | Front door speaker | - |
| 19 | A-pillar speaker | LF shown, RF similar |

**Overview**

The audio system consists of the following:

**Main Components**

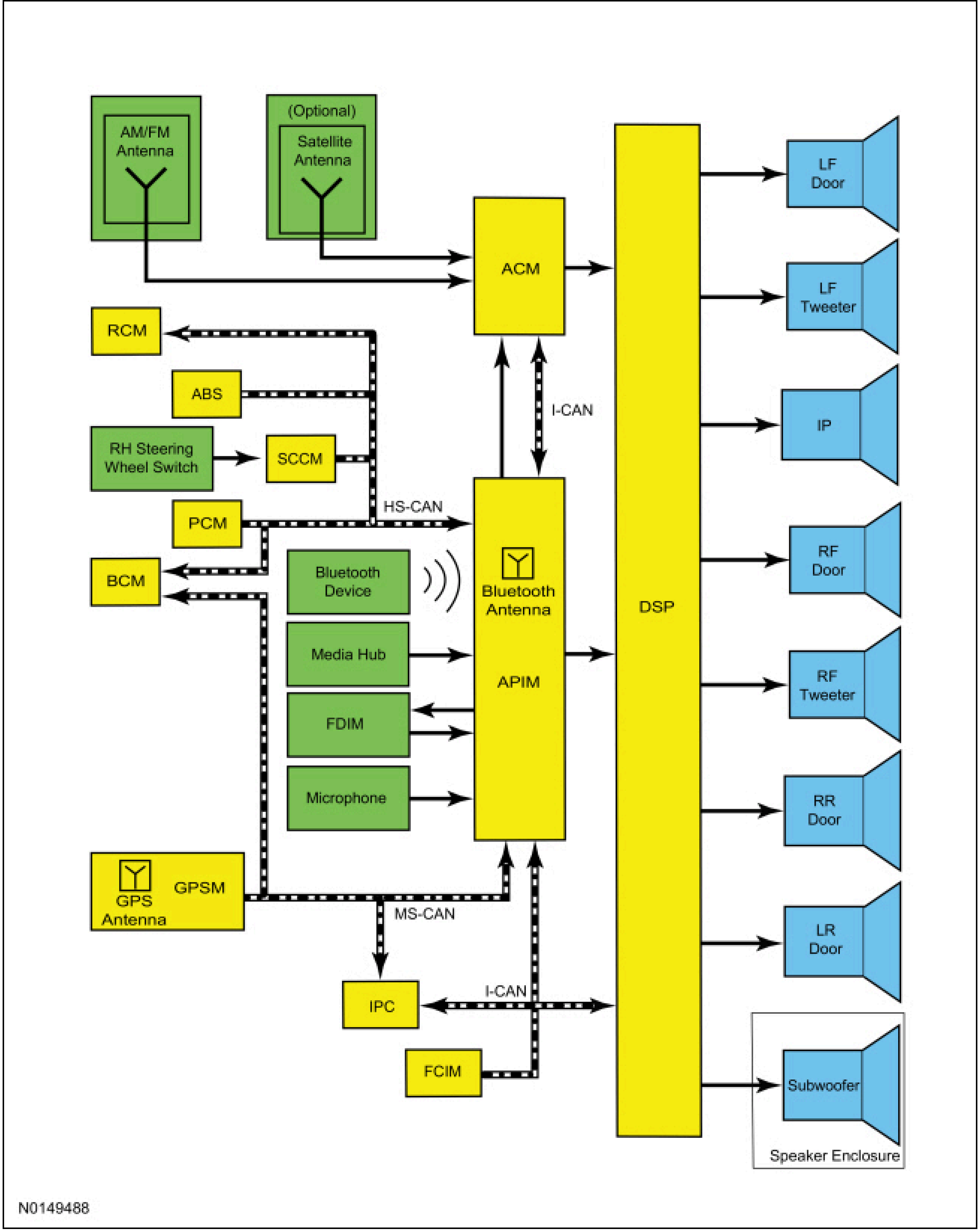
- 8 speakers
- 8-inch (203 mm) FDIM touchscreen
- AM/FM single CD ACM
- APIM
- Audio DSP module
- FCIM
- GPSM
- Media hub
- Redundant steering wheel controls
- SYNC® microphone

**Main Features**

- Optional HD Radio®
- Optional navigation
- Optional satellite radio
- SYNC®

**System Operation**

**System Diagram**




## Network Message Chart

**NOTE:** *A missing network message is indicated by a U-code DTC . When a missing message DTC (U-code) sets, it is important to look for symptoms present in the audio system and throughout the vehicle, and to review the complete message list to determine which other modules rely on the same message. Refer to CAN Module Communication Message Chart in Section 418-00. When the other modules have been identified, run the self-test for those modules. If the same message is missing from those modules, the identical, or similar lost communication DTCs may be set in those modules. Confirmation of missing messages common to multiple modules can indicate that the originating module is the source of the concern, or that the communication network may be at fault.*

*It is very important to understand the following:*

- *Where the input originates*
- *All of the information necessary for a feature to operate*
- *Which module(s) receive(s) the input or command message*
- *Which module controls the output of the feature*
- *If the module that receives the input controls the output of the feature, or whether it outputs a message over the CAN to another module.*

**Module Network Input Messages - ACM**

| Broadcast Message | Originating Module | Message Purpose |
|---|---|---|
| FCIM set volume | FCIM | Used to set the ACM volume from the FCIM . |
| Media player display request | APIM | Used to display detailed folder and track information on the FDIM from the ACM for the different audio system operating modes when the appropriate FCIM button is pressed. |
| SYNC® alerts | APIM | Used for text message, news, sports, weather, satellite radio artist/title favorites (if activated), and traffic alert notifications. |

**Module Network Input Messages - APIM**

| Broadcast Message | Originating Module | Message Purpose |
|---|---|---|
| ACM radio functions | ACM | Used to display the ACM radio functions such as seek, tune and AM/FM stations on the FDIM . |
| Airbag status | RCM | Used to monitor airbag deployment status for 911 Assist®. |
| Audio source | ACM | Used to display the selected audio source on the FDIM . |
| DSP volume status | Audio DSP module | Used to display the audio DSP module volume on the FDIM . |
| CD load/eject | ACM | Used to display the loaded or ejected CD status on the FDIM . |
| Date and time | IPC | Used to display the current date and time on the FDIM . |
| Door ajar status | BCM | Used to determine the door ajar status for the accessory delay feature. |
| FCIM button press | FCIM | Used to indicate when a button is pressed on the FCIM so the audio system can make the desired setting change. |
| Front seat belt latched status | RCM | Used to mute speaker output so the Belt-Minder® tone can be more easily heard. This message only applies to vehicles with a MyKey® enabled and in use. The audio system is muted until the safety belt(s) are buckled. |
| Ignition key type | BCM | Used to determine if a MyKey® is enabled and in use. |
| Ignition status | BCM | Used to indicate the ignition state (OFF, ACCESSORY, RUN, and START) required for APIM operating modes and fault reporting. |

| Broadcast Message | Originating Module | Message Purpose |
|---|---|---|
| Illumination dimming level | BCM | Used to control the backlight intensity of the FDIM based on the dimmer switch input. |
| Language selection | IPC | Used to display information on the FDIM in the selected language. |
| Navigation rolling wheel count | ABS module | Used to provide more accurate vehicle position tracking when the GPS signal is temporarily unavailable. |
| Seat belt buckle status | RCM | Used to mute speaker output so the Belt-Minder® tone can be more easily heard. This message only applies to vehicles with a MyKey® enabled and in use. The audio system is muted until the safety belt(s) are buckled. |
| Steering wheel switch media | SCCM | Used to set the audio system listening mode (AM/FM, CD, etc.) from the steering wheel switch. |
| Steering wheel switch seek left | SCCM | Used to access the next strong station to the left, to restart an audio track, to access the previous satellite radio station preset (if equipped), or the previous audio track from the steering wheel switch. |
| Steering wheel swtich seek right | SCCM | Used to access the next strong station to the right, to access the next satellite radio station preset (if equipped), or to access the next audio track from the steering wheel switch. |
| Steering wheel switch voice | SCCM | Used to access or exit the voice command feature from the steering wheel switch. |
| Steering wheel switch volume down | SCCM | Used to decrease the audio system volume from the steering wheel switch. |
| Steering wheel switch volume up | SCCM | Used to increase the audio system volume from the steering wheel switch. |

**Module Network Input Messages - Audio DSP Module**

| Broadcast Message | Originating Module | Message Purpose |
|---|---|---|
| Belt-Minder® audio mute | IPC | Used to mute speaker output so the Belt-Minder® tone can be more easily heard. This message only applies to vehicles with a MyKey® enabled and in use. The audio system is muted until the safety belt(s) are buckled. |
| Ignition status | BCM | Used to indicate the ignition state (OFF, ACCESSORY, RUN, and START) required for audio DSP module operating modes and fault reporting. |
| MyKey® volume limit | IPC | Used to limit the maximum audio system volume when a MyKey® restricted key is in use. |
| Navigation audio settings | APIM | Used to set the navigation command volume. |

**Module Network Input Messages - FCIM**

| Broadcast Message | Originating Module | Message Purpose |
|---|---|---|
| Door ajar status | BCM | Used to determine the door ajar status for the accessory delay feature. |
| Ignition status | BCM | Used to indicate the ignition state (OFF, ACCESSORY, RUN, and START) required for FCIM operating modes and fault reporting. |
| Illumination dimming level | IPC | Used to control the backlight intensity of the FCIM based on the dimmer switch input. |

**Module Network Input Messages - GPSM**

| Broadcast Message | Originating Module | Message Purpose |
|---|---|---|
| Ignition switch position | BCM | Used to indicate the ignition state (OFF, ACCESSORY, RUN, and START) required for GPSM operating modes and fault reporting. |
| Selector lever (PRNDL) status | BCM | Used to verify the gear lever position for vehicle direction data. |

**Accessory Delay Feature**

The accessory delay feature allows the audio system to be operated for a preset period of time after the ignition is turned off and a front door has not been opened.

**AM/FM Radio**

The AM/FM fender-mounted antenna receives radio waves and sends them through the AM/FM antenna coaxial cable to the ACM . The radio waves are converted into fluctuating AC voltage and sent as left and right analog audio signals to the audio DSP module. The audio DSP module processes the audio signals and sends them to the speakers as fluctuating AC voltage.

**APIM Programming**

The APIM requires specific programming procedures for correct operation. APIM programming is required when:

- The APIM is replaced
- Directed by a TSB. Updates to resolve SYNC® performance concerns can only be performed when directed by a TSB.
- Feature upgrades are requested. Occasional feature upgrades and/or new applications can be programmed when they become available. This type of programming is available upon customer request and expense.

There are 3 types of APIM programming available:

- Module Replacement — Used when the APIM has been replaced. This type of programming performs the required provisioning (restores software for newly installed hardware).
- Standard — Used to update the APIM and any currently installed Applications to the latest software level.
- Custom — Used to show all the available software options (including applications that may not have been installed previously) that can be programmed into the APIM . Use this selection when upgrades become available or as directed by a TSB.

There are 3 types of software installation methods:

- Full Flash (Force CIP ) — A CIP program is the entire software package for the APIM . Performing a Full Flash erases the existing APIM software and installs the complete program. A Full Flash should only be performed if a Service Pack is not available. If a Full Flash is performed, all applications must be re-installed. Performing a Full Flash by selecting Force CIP requires additional programming time and is not necessary unless directed by a TSB.
- Service Pack — A Service Pack update only provides the necessary software to update the existing APIM program to a new level. For example, a Service Pack changes the software program from an 'AA' to an 'AB' version level. It does not delete or overwrite the existing APIM software. A Service Pack is a much smaller program than a Full Flash and requires less downloading and programming time. A Service Pack software installation should always be performed instead of a Full Flash, if possible.
- Applications — Applications are special software programs that may be installed on the APIM . Applications necessary for each vehicle are determined by the vehicle build information stored in the Ford online database. Selecting individual applications for installation may not be necessary every time the APIM is programmed.

**NOTE:** *VIP programming is not selectable because the VIP is configured automatically during programming.*

There are some general programming guidelines that are applicable to multiple types of programming:

- Full Flash/Force CIP
  - The media hub must be removed for programming.
  - The diagnostic scan tool and SYNC® Re-Flash Kit (or equivalent), are required to complete the flash procedure.
  - All previously paired phones are deleted from the SYNC® system.
  - The file needs to be downloaded from the Ford online database. Allow for extra time in order to complete the download. The compressed files are between 200 and 400 megabytes (MB), and once downloaded, are extracted to the full size of approximately 2 gigabytes (GB).
  - Once diagnostic scan tool programming is complete, applications need to be installed with the USB flash drive. This should be accomplished by reinstalling the media hub and connecting the USB flash drive to the media hub.
- Service Pack and Applications
  - A USB flash drive with a minimum 1 gigabyte (GB) storage capacity is required.
  - The file is downloaded from the Ford online database onto the flash drive. The files are then transferred via the flash drive to the APIM through the vehicle USB port.

To perform module replacement programming, refer to Accessory Protocol Interface Module (APIM) Module Replacement Programming

To perform standard programming, refer to Accessory Protocol Interface Module (APIM) Standard Programming

To perform custom programming, refer to Accessory Protocol Interface Module (APIM) Custom Programming

**Audible Prompts**

The APIM receives stereo inputs from connected mobile phones and mono inputs from the SYNC® microphone.

Stereo and mono inputs, voice prompts, the TTS feature, ringtones, and any audio received through a connected mobile phone are transmitted to the ACM as stereo or mono sound.

The TTS feature speaks information so that it does not have to be read from the display.

**Audio Extended Play**

This feature allows the audio system to operate with the ignition OFF, regardless of door latch position. When the power button on the FCIM is pressed, audio system functionality is enabled and remains active for a period of 20-60 minutes, depending on configuration settings. To turn the audio system off, the power button on the FCIM is pressed. Audio extended play will not function if the battery load shed is active.

**Battery Load Shed**

The BCM uses the battery current sensor to monitor the battery state of charge. The battery current sensor is attached to the battery ground cable. With the vehicle OFF and the ignition in ACC or ON, a load shed message is sent over the CAN when the BCM determines that the battery state of charge is below 40%, 45 minutes have elapsed, or 10% of the charge has been drained. This message turns off the audio system to save the remaining battery charge. Under this condition, SYS OFF TO SAVE BATT is displayed on the centerstack infotainment display to notify the driver that battery protection actions are active. To clear the load shed state, start the vehicle.

**Bluetooth Mode**

Bluetooth is a secure, short-range radio frequency that allows devices to communicate wirelessly through radio waves. The operating range of a Bluetooth signal is a maximum of 9.75 m (32 feet).

The Bluetooth interface can accommodate both Bluetooth-enabled mobile phones and Bluetooth-enabled media devices. The SYNC® system allows interaction with several types of customer Bluetooth devices, including mobile phones and media devices. The APIM contains an on-board Bluetooth chipset, which enables certain wireless devices to interact with the system. Any Bluetooth device used with the SYNC® system must first be paired with the system before it is operational.

Only one Bluetooth phone and one Bluetooth media device can be connected to the system at any one time. If an additional device of either type is paired with the system and made active, the APIM disconnects any active connection and establishes a connection with the new device.

When a new Bluetooth device is added, the APIM and the Bluetooth device must be paired together. Most Bluetooth devices can pair with the SYNC® system, although functionality may vary. To determine if a Bluetooth device is supported, verify the customer device is on the compatibility list for the current APIM software level.

Pairing a Bluetooth device is accomplished through the "Add Device" selection of the phone menu. When pairing a device, the SYNC® system generates a unique PIN that must be entered on the Bluetooth device in order for the pairing process to be successful. There are also some device-specific actions that must take place. For information on the pairing process, refer to the Owner's Literature.

It is important to understand that not all mobile phones have the same level of features when interacting with the SYNC® system. For a list of compatible phones, Refer to SyncMyRide website.

**Clock Operation**

The APIM broadcasts a date/time status through CAN traffic messages for other modules to use whenever the CAN bus is awake and transmitting information. A default date/time value may be used by modules on the network if the date/time information is not available from the APIM.

**Composite Audio-Video Input Mode**

The composite audio and video inputs consist of the RCA jack connections that are used for playing video or audio content from an external device (such as a gaming system). The media hub contains an internal composite audio/video translator connected to the external RCA jacks, which outputs video and audio signals to the APIM via a shielded audio and video harness.

**DSP**

The Sony® audio system provides theater quality sound in the vehicle. Audio signals are sent from the ACM to the audio DSP module, where the signals are processed and sent to the speakers.

The audio DSP module determines the correct audio environment based on the audio signals and network messages it receives from other audio system modules. These messages include such information as fade/balance and speed compensated volume settings.

**HD Radio® (If Equipped)**

HD Radio® is a free to the public digital radio broadcast. HD Radio® channels duplicate certain analog FM channels, and some subdivide (multicast) the digital signal into multiple channels that do not have analog equivalents.

The AM/FM fender-mounted antenna receives radio waves containing data (station call letters, artist names, song titles, etc.) and digital HD Radio® audio. The radio waves are sent through the AM/FM antenna coaxial cable to the ACM where the data is separated from the audio. The data is sent to be displayed on the centerstack infotainment display, and the audio is converted into fluctuating AC voltage and sent as left and right analog audio signals to the audio DSP module. The audio DSP module processes the audio signals and sends them to the speakers as fluctuating AC voltage.

**MyKey® Audio Operation**

**When a MyKey® is in use, the following restrictions are activated:**

- The audio system is muted until the safety belts are buckled when the Belt-Minder® feature is activated. The MyKey® "BUCKLE UP TO UNMUTE AUDIO" is displayed in the information display.
  - This is a standard setting and cannot be changed.
- Satellite radio adult content stations are blocked by being muted.
- If "Always On" is selected in the MyKey® menu, 911 Assist® and the Do Not Disturb features are enabled and cannot be disabled.
- The audio system volume is limited to 45%. In an attempt to exceed the limited volume, the MyKey® VOLUME LIMITED message is displayed in the centerstack infotainment display.
  - This is an optional setting.

Refer to the Owner's Literature for more information on MyKey®.

**Navigation**

The navigation system guides the user to a pre-entered destination. Map data is read from the map data SD card plugged into the media hub. The APIM calculates route information based on GPS data received by the GPSM . The APIM also uses vehicle speed and transmission gear selected signals received through the network to detect vehicle speed and direction, resulting in more accurate navigation tracking. The navigation display is shown on the FDIM . Installing different size tires other than what is indicated on the vehicle placard may cause the navigation system to be inaccurate.

The voice recognition system allows the user to interface with the system without using the touchscreen. A microphone located in the headliner provides a direct input to the APIM .

**Satellite Radio (If Equipped)**

Satellite radio is a subscription-based service.

The satellite radio roof-mounted antenna receives signals containing data (station call letters, artist names, song titles, etc.) and digital satellite radio audio. These signals are sent through the satellite radio antenna coaxial cable to the satellite radio receiver built into the ACM . The ACM separates the data from the audio. The data is sent to be displayed on the centerstack infotainment display, and the audio is converted into fluctuating AC voltage and sent as left and right analog audio signals to the audio DSP module. The audio DSP module processes the audio signals and sends them to the speakers as fluctuating AC voltage.

**SD Card Mode**

The SD card slot can be used for playing audio content that is stored on a SD card.

Image files can be saved to the APIM from a SD card in order to be used as wallpaper backgrounds displayed on the FDIM .

SD card data is transmitted to the APIM through the USB cable.

**SIRIUS Data Services - If Equipped**

SIRIUS Data Services is a subscription-based service and requires a separate subscription from the satellite radio.

The combination HD /satellite radio roof-mounted antenna receives digital signals and sends them to the satellite radio receiver through the satellite radio antenna coaxial cable. The satellite radio receiver is built into the ACM . The signals are then sent to the APIM , and then to the FDIM touchscreen to be displayed.

**SIRIUS Data Services allows the customer to receive updates for:**

- Fuel Prices
- Movies (movie listings, start times, ratings)
- Sports (in game and final scores, weekly schedules)
- Stocks
- Ski Conditions
- Traffic (traffic speed, accidents, construction, road closings)
- Weather (current weather, forecasts, radar images)

Some services may be available separately while others are only available as packages.

**Speed Compensated Volume**

The audio DSP module adjusts the audio system volume based on the VSS signal to compensate for speed and wind noise.

When a MyKey® is in use and the MyKey® radio volume limiter is enabled, the speed compensated volume is disabled.

**Steering Wheel Switch Functions**

The RH 8-way steering wheel switches operate the volume +/-, seek +/-, and the audio system SYNC® functions (media, voice, and phone).

**SYNC® System**

The SYNC® system is a hands-free communication and entertainment system that supports many features. All features are not available with every phone.

**The SYNC® system provides the ability to:**

- Connect media devices (such as an iPod® or USB device) in order to play audio files.
- Control the entertainment system, climate control system, ambient lighting system, and several other vehicle systems using the touchscreen or voice controls.
- Initiate an emergency call (911 Assist®) when the airbags deploy.
- Play media files via a paired Bluetooth-enabled audio device.
- Produce vehicle health reports.
- Send and receive phone calls via a paired Bluetooth-enabled phone.
- Send and receive text messages via a paired Bluetooth-enabled phone.
- Wi-Fi signal receiver for in-car Internet access (requires customer-supplied USB modem).

**SYNC® Services**

SYNC® Services is a subscription-based service. For SYNC® Services subscription information, refer to SyncMyRide.

The customer's mobile phone is used to access and download the SYNC® services. SYNC® services information received by the customer's mobile device is sent to the APIM .

The GPS antenna, integral to the GPSM , is used to detect the location and direction of the vehicle.

**SYNC® services includes the following features:**

- Business search
- Personalized text alerts
- Real-time traffic updates
- Turn-by-turn directions

**SYNC® services allows the customer to receive updates for:**

- Entertainment
- News
- Sports
- Stocks

- Travel
- Weather

### USB Mode

The USB port can be used for connecting a media device (such as an iPod®) with the device's available cable, or for directly plugging in a portable mass storage device, such as a thumb drive. When playing media files stored on a mass storage device, the SYNC® system only plays files that do not have DRM protection. The USB port can also be used for uploading vehicle application upgrades. The USB port is powered by the APIM , so no external power source is needed to power a device plugged into the USB port if the device supports this feature.

In addition to audio information, metadata (information such as artist, album title, song title, and genre) may also be sent to the APIM from a device plugged into the USB port. The APIM uses the metadata to create indexes that can be used to sort for particular music, based on customer preference. Not all USB devices can send metadata to the APIM . When a new media device is connected to the SYNC® system, the APIM automatically indexes the information. This may take several minutes (depending on the amount of data on the device), and is considered normal operation. When a device that was previously connected to the SYNC® system is reconnected, the APIM updates the index (rather than creating a new one), which reduces the amount of time needed to index the device.

Image files can be saved to the APIM from a USB device. These images can be displayed on the FDIM as wallpaper backgrounds.

### Voice Recognition

Voice recognition is used for many vehicle functions, including audio system and climate controls. The microphone relays the microphone input to the APIM through dedicated wiring. The TTS and voice prompt features speak certain text information and interaction requests to minimize driver distraction by having to look at the audio system display while driving. The ringtone alerts the driver to an incoming call. The microphone is also used to detect outgoing audio during a phone call.

Audible prompts can range from a simple tone to more elaborate spoken text, based on the customer setting. When interaction mode is set to standard, detailed guidance is provided. When interaction mode is set to advanced, most prompts are tones only and minimal audible guidance is provided. Refer to the Owner's Literature for further information on voice interaction.

The audio signals for the TTS and voice prompt features, ringtones, and audio from the outside device during a phone call, are sent from the APIM to the ACM .

## Component Description

### ACM

The ACM can be operated with the ignition in RUN, ACC, or OFF.

The ACM receives radio waves containing audio from the antenna coaxial cable(s) and converts them into fluctuating AC voltage. The ACM receives audio signals from the APIM in the form of fluctuating AC voltage. The ACM processes these inputs and sends left and right analog audio signals to the audio DSP module as fluctuating AC voltage. The audio DSP module processes the audio signals and sends them to the speakers as fluctuating AC voltage.

If equipped with satellite radio or HD Radio®, the ACM will also receive radio waves containing data through the antenna coaxial cable(s). The data is sent to be displayed on the centerstack infotainment display.

The ACM requires PMI when it is replaced.

### Antenna

The AM/FM fender-mounted antenna receives radio waves containing audio and sends them through the AM/FM antenna coaxial cable to the ACM . The antenna mast is used to amplify the AM/FM radio waves to improve reception.

If equipped with HD Radio®, the AM/FM fender-mounted antenna receives radio waves containing data and digital audio and sends them through the AM/FM antenna coaxial cable to the ACM .

If equipped with satellite radio, the satellite radio roof-mounted antenna receives signals containing data and digital audio. These signals are sent through the satellite radio antenna coaxial cable to the satellite radio receiver built into the ACM .

**APIM**

Audio signals received by the APIM are sent to the ACM as fluctuating AC voltage.

The APIM consists of 2 internal modules: the CIP and the VIP . The modules are not replaceable individually, but can be flashed independently, if required.

The CIP interfaces with all of the inputs to the APIM . It contains an analog-to-digital-to analog converter, as well as the Bluetooth chipset. Any Application upgrades that are available to the consumer are loaded directly to the CIP through the USB port.

The VIP provides an interface between the CIP and the vehicle. Its main functions are controlling the APIM power management and translating inbound and outbound CAN signals. In addition, the VIP queries the modules on the network to retrieve any DTCs when a vehicle health report is requested.

**Audio DSP Module**

The audio DSP module enhances the input audio signal sound quality from the ACM .

Audio signals received by the audio DSP module are sent to the audio system speakers as fluctuating AC voltage.

The audio DSP module receives audio signals from the ACM in the form of fluctuating AC voltage. The audio DSP module processes these inputs and sends left and right analog audio signals to the speakers as fluctuating AC voltage.

The audio DSP module requires PMI when it is replaced.

**FCIM**

The FCIM can be operated with the ignition in RUN, ACC, or OFF.

The FCIM audio system buttons provide one of the methods in which the customer interacts with the infotainment system.

The FCIM is separate from the ACM .

**FDIM**

The FDIM can be operated with the ignition in RUN, ACC, or OFF.

The FDIM touchscreen provides one of the methods in which the customer interacts with the infotainment system.

The FDIM plugs directly into the APIM , and therefore does not communicate directly on any network, and no external circuits are connected.

**Global Positioning System Module (GPSM)**

The Global Positioning System Module (GPSM) provides vehicle location for real-time traffic reports and re-routing, and for identifying vehicle location in the event of a collision.

For vehicles with navigation, the Global Positioning System Module (GPSM) acts as the antenna for the navigation system.

Vehicle location information is broadcast from the Global Positioning System Module (GPSM) to the APIM .

**Media Hub**

The media hub allows for various audio and video devices to play content through the vehicle speakers and to be viewed in the FDIM .

The media hub receives inputs from:

- 2 USB ports
- 1 Secure Digital (SD) card slot
- 1 set of component RCA jacks

**Steering Wheel Switches**

The RH steering wheel switches contain series of resistors. Each steering wheel switch function corresponds with a specific resistance value within the switch.

The SCCM sends a 5-volt reference voltage to the RH steering wheel switch on the input circuits. When a switch is pressed, the voltage is routed through the corresponding resistor(s), and then to ground through the SCCM . The SCCM monitors the resultant voltage drop to determine which switch was pressed. The voltage drop varies depending upon the resistance of the specific switch pressed.

The SCCM sends a network message to the APIM for the SYNC® switches and to the ACM for the redundant audio controls.

**Subwoofer Speaker**

The subwoofer speaker is contained within an enclosure and recieves audio signals from the audio DSP module.

**SYNC® Microphone**

The SYNC® microphone receives SYNC® system voice commands and outgoing audio during a phone call. These signals are sent to the APIM .